

Module 06: Mail Access Protocols: POP3 vs IMAP4

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MESSAGE ACCESS PROTOCOLS: POP3 vs IMAP4 ARCHITECTURE

Mail Retrieval Mechanics, Delete vs Keep Mode, Server-Side Folder Synchronization & Bandwidth

POP3 (Post Office Protocol v3 — RFC 1939)

Port: TCP Port 110 (Cleartext) | Port 995 (POP3S over SSL) • **Core Philosophy:** Simple offline download-and-read model. • **Delete Mode:** Mailbox mail deleted from server upon download. • **Keep Mode:** Copies kept on server, but states not synchronized. • **Single Device Bound:** Multi-device usage causes state divergence. • **No Folder Hierarchy:** Cannot create or organize server-side folders. • **Full Download Only:** Must download full message including attachments. Commands: USER, PASS, STAT, LIST, RETR, DELE, QUIT IMAP4 (Internet Message Access Protocol v4 — RFC 3501) • **Port:** TCP Port 143 (Cleartext) | Port 993 (IMAPS over SSL) • **Core Philosophy:** Dynamic client-server synchronized database. • **Multi-Device Sync:** Phone, laptop & webmail see identical mailbox state. • **Message Flags:** \Seen, \Answered, \Flagged, \Deleted tracked centrally. • **Server Folders:** Create, rename, delete hierarchical folders on server. • **Partial Fetch:** Download headers or preview without 20MB attachments. • **Server-Side Search:** Search keywords without downloading all emails. Commands: LOGIN, SELECT, FETCH, STORE, COPY, SEARCH, LOGOUT MASTER COMPARATIVE MATRIX: POP3 vs IMAP4 1. Protocol Role & Ports • **Type:** Both are Message Access (PULL) • **POP3:** Port 110 (Clear) / 995 (SSL) • **IMAP4:** Port 143 (Clear) / 993 (SSL) • **SMTP Role:** Pushes mail; cannot pull. 2. Storage & Sync Mechanics • **POP3:** Local client machine storage. • **IMAP4:** Central mail server storage. • **Read Status:** POP3 cannot sync read tags; IMAP4 syncs across all logged-in devices. 3. Bandwidth & Resource Efficiency • **POP3:** Requires full file download. • **IMAP4:** Headers-only preview saves data. • **Server Load:** POP3 has very low RAM churn; IMAP4 requires robust high-RAM servers.

1. Message Access Paradigm (PULL vs PUSH)

While SMTP handles inter-server PUSH operations, client devices require PULL protocols (POP3 or IMAP4) to retrieve messages from intermediate mailbox buffers.

Architectural Divergence

POP3 views the mail server as a temporary post office drop box; IMAP4 views the server as a perpetual, synchronized cloud database.

2. POP3 Operational Modes

- **Delete Mode:** Mail is transferred to client hard drive and immediately purged from server spool.
- **Keep Mode:** Messages persist on the server following retrieval, but message state changes remain strictly local.

3. Master Comparative Breakdown: POP3 vs IMAP4

Criterion	POP3 (RFC 1939)	IMAP4 (RFC 3501)
Well-Known Ports	Port 110 (Plain) / 995 (SSL)	Port 143 (Plain) / 993 (SSL)
Primary Storage Target	Client host disk	Server-side database storage

Multi-Device Consistency	Fails across multiple concurrent devices	Maintains real-time status synchronization
Partial Retrieval	Requires complete segment download	Allows isolated header and MIME part preview
Server Complexity	Minimal memory and disk utilization	Resource intensive (CPU, indexing, disk quota)